

# Macroeconomics #8

## Review

Quang-Thanh Tran

RCPD, Tohoku University

CASED Pre-Master Macroeconomics  
July 7, 2026

## Starting point

- ① You have a research question, an inquiry
- ② You build a model to see what happens if I have this feature
- ③ You analyze the model to tell and sell a story or a policy

## The flow

- ① Write your model, solve each actor's problem.
- ② Define your equilibrium and bring all the actors to the markets.
- ③ Solve for the equilibrium, clear the markets.
- ④ Find the dynamics of the equilibrium and the steady state.
- ⑤ Pick what story to tell
- ⑥ You can find empirical evidence to support the story, or alter the model to make it policy-relevant and sell a policy story.

# The Ramsey–Cass–Koopmans Framework

# Ramsey Model: Setup

**Representative household** maximizes

$$V_0 = \sum_{t=0}^{\infty} \beta^t U(c_t), \quad \beta \in (0, 1),$$

subject to the resource constraint

$$c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t, \quad k_0 \text{ given.}$$

**CRRA (constant relative risk aversion) utility:**

$$U(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}, \quad \sigma > 0.$$

- $\sigma$  is the coefficient of relative risk aversion.
- Also governs the willingness to substitute consumption across time.

# Optimality: Euler Equation

From the Lagrangian

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \left\{ \frac{c_t^{1-\sigma} - 1}{1-\sigma} + \lambda_t [f(k_t) + (1-\delta)k_t - c_t - k_{t+1}] \right\}.$$

FOCs:

$$\begin{aligned} (c_t) : \quad & c_t^{-\sigma} = \lambda_t \\ (k_{t+1}) : \quad & \lambda_t = \beta \lambda_{t+1} [f'(k_{t+1}) + 1 - \delta] \end{aligned}$$

**Euler equation:**

$$c_t^{-\sigma} = \beta c_{t+1}^{-\sigma} [f'(k_{t+1}) + 1 - \delta].$$

**Consumption growth:**

$$\frac{c_{t+1}}{c_t} = \left( \beta [f'(k_{t+1}) + 1 - \delta] \right)^{1/\sigma}.$$

# Intertemporal Elasticity of Substitution (IES)

**Question:** If the return to saving increases, how much will households shift consumption from today to tomorrow?

The **Intertemporal Elasticity of Substitution (IES)** measures the willingness to substitute consumption across time.

$$\text{IES} \equiv -\frac{U'(c)}{cU''(c)}$$

It measures the *percentage change in the consumption growth ratio* induced by a one-percent change in the intertemporal price (the gross real interest rate).

**For CRRA preferences**

$$U(c) = \frac{c^{1-\sigma} - 1}{1-\sigma},$$

$$U'(c) = c^{-\sigma}, \quad U''(c) = -\sigma c^{-\sigma-1},$$

$$\text{IES} = \frac{1}{\sigma}.$$

Thus, the coefficient of relative risk aversion ( $\sigma$ ) is also the inverse of the willingness to substitute consumption over time.

## Connection with the Euler equation

$$\frac{c_{t+1}}{c_t} = [\beta(1 + f'(k_{t+1}) - \delta)]^{1/\sigma}.$$

Taking logs,

$$\Delta \ln c_{t+1} = \frac{1}{\sigma} \left[ \ln(r_{t+1}) - \ln\left(\frac{1}{\beta}\right) \right]$$

where  $r_{t+1} = 1 + f'(k_{t+1}) - \delta$  is the return on savings.

- **High IES** ( $\sigma$  small): households readily postpone consumption. A higher real interest rate causes a large increase in saving and faster consumption growth.
- **Low IES** ( $\sigma$  large): households dislike changing consumption over time. Even if the return to saving rises, consumption changes little.

# Steady State: Modified Golden Rule

Set  $c_{t+1} = c_t = c^{ss}$ ,  $k_{t+1} = k_t = k^{ss}$  in the Euler equation:

$$1 = \beta[f'(k^{ss}) + 1 - \delta].$$

$$f'(k^{ss}) = \frac{1}{\beta} - 1 + \delta \equiv \delta + \theta, \quad \theta \equiv \frac{1 - \beta}{\beta}.$$

- $k^{ss}$  is determined *solely* by impatience ( $\beta$ ) and technology ( $\delta, f$ ).
- The utility parameter  $\sigma$  does *not* affect the steady-state capital stock.
- $\sigma$  shapes only the **transition path** (speed of convergence).

# Stone–Geary Preferences: Motivation

The standard CRRA model assumes that the IES is the same for rich and poor.

However, if subsistence consumption exists, households may not be able to save (defer consumption) when their consumption level near subsistence.

- Stone–Geary utility introduces a **subsistence level**  $\bar{c} \geq 0$ .
- The household must consume at least  $\bar{c}$  to obtain any utility.

**Utility function:**

$$U(c) = \frac{(c - \bar{c})^{1-\sigma} - 1}{1 - \sigma}, \quad \sigma > 0, \bar{c} \geq 0.$$

- When  $\bar{c} = 0$ , we recover the standard CRRA case.
- Define **effective consumption**:  $\tilde{c}_t \equiv c_t - \bar{c}$ .

## Euler equation:

$$\tilde{c}_t^{-\sigma} = \beta \tilde{c}_{t+1}^{-\sigma} [f'(k_{t+1}) + 1 - \delta].$$

## IES:

$$\text{IES}(c) = -\frac{U'(c)}{cU''(c)} = \frac{c - \bar{c}}{\sigma c} = \frac{1}{\sigma} \left(1 - \frac{\bar{c}}{c}\right).$$

- **IES is increasing in  $c$**  (wealthier households are more willing to substitute).
- As  $c \rightarrow \bar{c}$  (near subsistence):  $\text{IES} \rightarrow 0$  — cannot defer consumption.
- As  $c \rightarrow \infty$ :  $\text{IES} \rightarrow 1/\sigma$  — converges to CRRA.

# Stone–Geary: Consumption Growth

The growth rate of **effective** consumption follows the CRRA law:

$$\frac{\tilde{c}_{t+1}}{\tilde{c}_t} = \left( \beta [f'(k_{t+1}) + 1 - \delta] \right)^{1/\sigma}.$$

The growth rate of **actual** consumption:

$$\frac{c_{t+1}}{c_t} = \frac{(\beta R_{t+1})^{1/\sigma} \tilde{c}_t + \bar{c}}{\tilde{c}_t + \bar{c}}.$$

## Intuition:

- Actual growth is a *weighted average* of the CRRA rate and 1.
- Weight on 1 is  $\bar{c}/c_t$  — large near subsistence, small far away.
- The subsistence requirement acts as a **floor**: even if returns are high, the household cannot defer too much consumption.

At  $c_{t+1} = c_t = c^{ss}$ :

$$f'(k^{ss}) = \frac{1}{\beta} - 1 + \delta.$$

- $k^{ss}$  is **unchanged** by  $\bar{c}$
- However, **feasibility** condition:  $c^{ss} = f(k^{ss}) - \delta k^{ss} \geq \bar{c}$ . If  $\bar{c}$  is too large, the steady state is *infeasible*.
- Transitional dynamics are richer: the speed of convergence depends on how far  $c_t$  is above  $\bar{c}$ , since the effective IES varies with  $c_t$ .

# The Overlapping-Generations Framework

# Longevity, Saving, and Fertility

## Environment:

- Time is discrete  $t = 0, 1, 2, \dots$ .  $N_t$  identical young agents born in each  $t$ .
- Each agent lives two periods, but survives to old age with probability  $\pi \in (0, 1]$ .
- Annuity markets exist: survivors receive  $R_{t+1}/\pi$  per unit saved.
- Population evolves as  $N_{t+1} = n_t N_t$ .

## Household preferences:

$$U = \ln c_t^y + \beta \pi \ln c_{t+1}^o + \gamma \ln n_t.$$

## Household budget constraints:

$$c_t^y + s_t = (1 - \phi n_t) w_t,$$

$$c_{t+1}^o = \frac{R_{t+1} s_t}{\pi}.$$

where  $\phi$  = time cost per child;  $w_t$  = wage;  $s_t$  = saving.

## Lagrangian:

$$\mathcal{L} = \ln c_t^y + \beta\pi \ln c_{t+1}^o + \gamma \ln n_t + \lambda \left[ (1 - \phi n_t)w_t - c_t^y - s_t \right].$$

## FOCs ( $c_t^y, s_t, n_t$ ):

$$\frac{1}{c_t^y} = \lambda, \quad \frac{\beta\pi}{s_t} = \lambda, \quad \frac{\gamma}{n_t} = \lambda\phi w_t.$$

## Optimal choices:

$$c_t^y = \frac{w_t}{1 + \beta\pi + \gamma}, \quad s_t = \frac{\beta\pi w_t}{1 + \beta\pi + \gamma}, \quad n_t = \frac{\gamma}{\phi(1 + \beta\pi + \gamma)}.$$

- Consumption and saving are proportional to the wage.
- Fertility is *independent* of the wage (income and substitution effects cancel).
- Higher survival  $\pi \Rightarrow$  higher saving (longer retirement), lower fertility.

# General Equilibrium: Production and Factor Prices

## Production (Cobb–Douglas):

$$Y_t = AK_t^\alpha L_t^{1-\alpha}, \quad 0 < \alpha < 1.$$

Labor  $L_t = N_t \times (1 - \phi n_t)$  (each young supplies one unit, net of child-rearing time). Per-worker output:  $y_t = Ak_t^\alpha$ ,  $k_t \equiv K_t/N_t$ . Capital fully depreciates after 1 period.

## Profit

$$\Pi_t = Y_t - w_t L_t - R_t K_t$$

**Competitive factor prices** obtained from profit maximization:

$$w_t = (1 - \alpha) A k_t^\alpha, \quad R_t = \alpha A k_t^{\alpha-1}.$$

## Capital accumulation:

$$\begin{cases} K_{t+1} = N_t s_t, \\ N_{t+1} = n_t N_t \end{cases} \implies k_{t+1} = \frac{s_t}{n_t}.$$

Each young agent saves  $s_t$ , which becomes next period's capital stock.

# Competitive Equilibrium: Definition

A **competitive equilibrium** is a sequence of allocations and prices

$$\{c_t^y, c_t^o, s_t, n_t, k_t, w_t, R_t\}_{t=0}^{\infty}$$

such that, **given** the initial capital stock  $k_0$ , the following three conditions hold at every date  $t$ :

① **Household optimality.**

$$c_t^y = \frac{w_t}{1 + \beta\pi + \gamma}, \quad s_t = \frac{\beta\pi w_t}{1 + \beta\pi + \gamma}, \quad n_t = \frac{\gamma}{\phi(1 + \beta\pi + \gamma)}.$$

② **Firms' profit maximization (factor prices).**

$$w_t = (1 - \alpha)A k_t^\alpha, \quad R_t = \alpha A k_t^{\alpha-1} + 1 - \delta.$$

③ **Market clearing (capital accumulation).**

$$k_{t+1} = \frac{s_t}{n_t}, \quad c_t^o = \frac{R_t}{\pi} s_{t-1}.$$

These three blocks together determine the entire time path of the economy from any initial  $k_0$ .

Substitute the optimal  $s_t$  and  $n_t$  into  $k_{t+1} = s_t/n_t$ :

$$k_{t+1} = \frac{\frac{\beta\pi w_t}{1 + \beta\pi + \gamma}}{\frac{\gamma}{\phi(1 + \beta\pi + \gamma)}} = \frac{\beta\pi\phi}{\gamma} w_t.$$

Using  $w_t = (1 - \alpha)A k_t^\alpha$ :

$$k_{t+1} = \frac{\beta\pi\phi(1 - \alpha)A}{\gamma} k_t^\alpha.$$

This is a **first-order nonlinear difference equation** in  $k_t$ .

- $0 < \alpha < 1 \Rightarrow$  the map  $k_t \mapsto k_{t+1}$  is concave and globally stable.
- The economy converges monotonically to a unique steady state.

Fertility dynamics:  $n_t$  is *constant* over time (not state-dependent):

$$n_t = \frac{\gamma}{\phi(1 + \beta\pi + \gamma)}, \quad \forall t.$$

# Steady State

Impose  $k_{t+1} = k_t = k^{ss}$ :

$$k^{ss} = \frac{\beta\pi\phi(1-\alpha)A}{\gamma} (k^{ss})^\alpha \implies k^{ss} = \left( \frac{\beta\pi\phi(1-\alpha)A}{\gamma} \right)^{\frac{1}{1-\alpha}}.$$

**Steady-state fertility:**

$$n^{ss} = \frac{\gamma}{\phi(1 + \beta\pi + \gamma)}.$$

# Comparative Statics: Longer Life Expectancy

Effect of higher survival probability  $\pi \uparrow$ :

$$\frac{\partial k^{ss}}{\partial \pi} > 0, \quad \frac{\partial n^{ss}}{\partial \pi} < 0.$$

- **Capital deepening:** longer life  $\Rightarrow$  more saving for retirement  $\Rightarrow$  higher  $k^{ss}$ .
- **Fertility decline:** longer life  $\Rightarrow$  higher marginal utility of old-age consumption  $\Rightarrow$  resources shift from children to saving.

**This is the demographic transition channel:**

$$\text{Longevity } \uparrow \Rightarrow \text{Saving } \uparrow, \text{ Fertility } \downarrow \Rightarrow k^{ss} \uparrow, y^{ss} \uparrow.$$

The model rationalizes why rising life expectancy is associated with both higher capital intensity and lower population growth.

# Empirical Evidence: Longevity, Income, and Fertility

